Information Collection Strategies Experiment

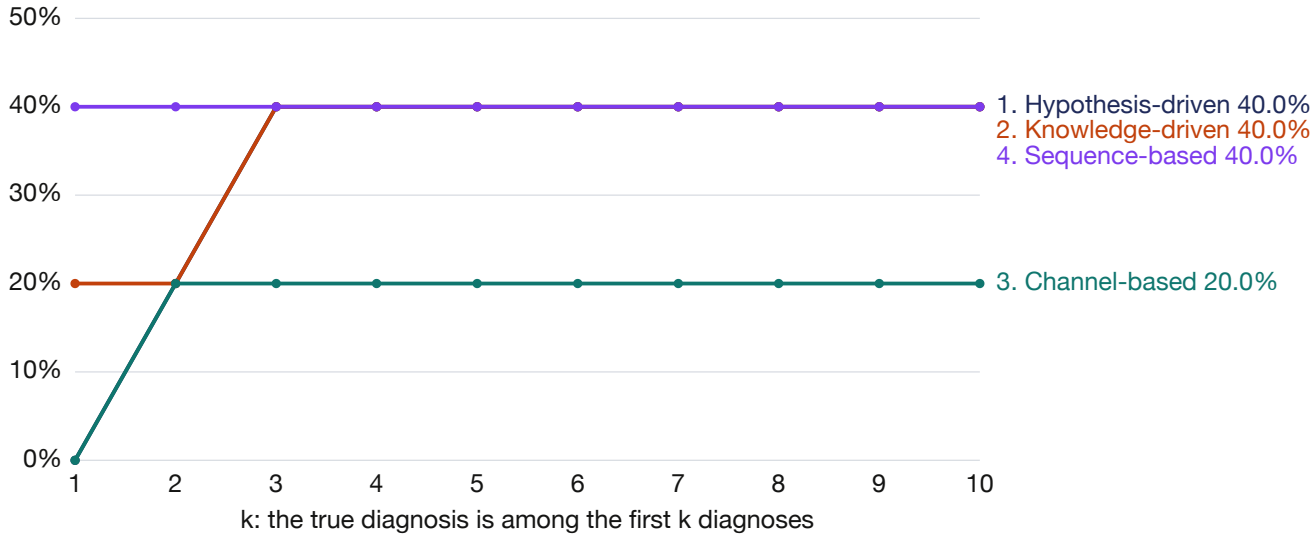
Diagnostic accuracy of four information collection strategies on 5 simulated cases

Run 2026-10-12T16:51:00+00:00 to 2026-10-12T16:53:27+00:00 · record `pilot_strategies_272_20261012T165100Z.json` · status completed · commit 35c54b4

Result

Strategy	Hit rate@1	Hit rate@3	Hit rate@5	Hit rate@10	nDCG@10	MRR	Cases
1. Hypothesis-driven	0.0%	**40.0%**	**40.0%**	**40.0%**	0.226	0.167	5
2. Knowledge-driven	20.0%	**40.0%**	**40.0%**	**40.0%**	0.300	0.267	5
3. Channel-based	0.0%	20.0%	20.0%	20.0%	0.126	0.100	5
4. Sequence-based	**40.0%**	**40.0%**	**40.0%**	**40.0%**	**0.400**	**0.400**	5

Bold marks the best value in a column. Hit rate@k: share of cases whose true diagnosis code is among the first k codes of the strategy's differential. nDCG@10 and MRR reward a true diagnosis placed higher. Every case counts in the denominator.



How sure the differences are

Strategy	Hit rate@1 (95% interval)	Hit rate@10 (95% interval)
1. Hypothesis-driven	0.0% (0.0% to 43.4%)	40.0% (11.8% to 76.9%)
2. Knowledge-driven	20.0% (3.6% to 62.4%)	40.0% (11.8% to 76.9%)
3. Channel-based	0.0% (0.0% to 43.4%)	20.0% (3.6% to 62.4%)
4. Sequence-based	40.0% (11.8% to 76.9%)	40.0% (11.8% to 76.9%)

Wilson intervals over the cases of this single run. The strategies ran on the same cases, so the paired comparison below is the sharper test.

Pair (first vs second)	@1: cases only the first / only the second hit	p	@10: only the first / only the second	p
1. Hypothesis-driven vs 2. Knowledge-driven	0 / 1	1.000	0 / 0	1.000
1. Hypothesis-driven vs 3. Channel-based	0 / 0	1.000	1 / 0	1.000
1. Hypothesis-driven vs 4. Sequence-based	0 / 2	0.500	1 / 1	1.000
2. Knowledge-driven vs 3. Channel-based	1 / 0	1.000	1 / 0	1.000
2. Knowledge-driven vs 4. Sequence-based	1 / 2	1.000	1 / 1	1.000
3. Channel-based vs 4. Sequence-based	0 / 2	0.500	0 / 1	1.000

Exact McNemar test on the same 5 cases. With 6 comparisons, a p-value below 0.008 (bold) holds after a Bonferroni correction; values between that and 0.05 are suggestive only.

What was run

- 5 cases from `cases/combined_272_whole_chunking.json`, each run with every strategy: 20 case runs.
- Each strategy could ask at most 4 questions and could stop earlier by its own rule. When it stopped, it gave a differential of up to 10 diagnoses with probabilities.
- Strategy model: anthropic/claude-haiku-5.5. Simulated environment model: openai/gpt-6-luna; literature sources europe_pmc, litsense.
- Coding: ICD-11 MMS 2024-01, first result of the flexible search for the diagnosis name. A prediction counts as a hit only when its code equals the true diagnosis code exactly.
- Total time 2 min; total cost $0.11 (strategies $0.05, environment $0.06).

How the strategies behaved

Strategy	Questions per case	Why runs stopped	Strategy LLM calls per case	Seconds per case	Strategy cost	Environment cost	Failed runs	No differential
1. Hypothesis-driven	4.00	max iterations 5	8.0	116	$0.02	$0.02	0	0
2. Knowledge-driven	4.00	max iterations 5	8.0	111	$0.01	$0.02	0	0
3. Channel-based	4.00	max iterations 5	10.0	98	$0.01	$0.02	0	0
4. Sequence-based	4.00	max iterations 5	8.0	94	$0.01	$0.01	0	0

Where the answers came from (questions)	Case study	Literature	Unanswerable	No answer (error)
1. Hypothesis-driven	6	13	1	0
2. Knowledge-driven	5	14	1	0
3. Channel-based	5	14	1	0
4. Sequence-based	6	13	1	0

Coding

Strategy	Predicted diagnoses	Without an ICD code	Cases with no coded prediction	Cases whose true diagnosis has no code
1. Hypothesis-driven	50	0	0	0
2. Knowledge-driven	50	0	0	0
3. Channel-based	50	0	0	0
4. Sequence-based	50	0	0	0

161 distinct diagnosis names were coded; 0 have no ICD-11 match. A case whose true diagnosis has no code cannot be hit by any strategy.

Reading these numbers

- Exact code match is strict: a correct diagnosis worded so that the search returns a neighbouring code (a parent, a sibling, or a code with a different extension) counts as a miss.
- One run per case and strategy. LLM sampling differs between runs, so small differences between strategies may not repeat.
- The stopping thresholds are starting values and were not tuned.
- Answers marked "literature" are synthesized by the simulated environment when the case text is silent; they are plausible, not facts about the patient.